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Lupin hull cellulose nanofiber aerogel preparation by supercritical CO_2 and freeze drying

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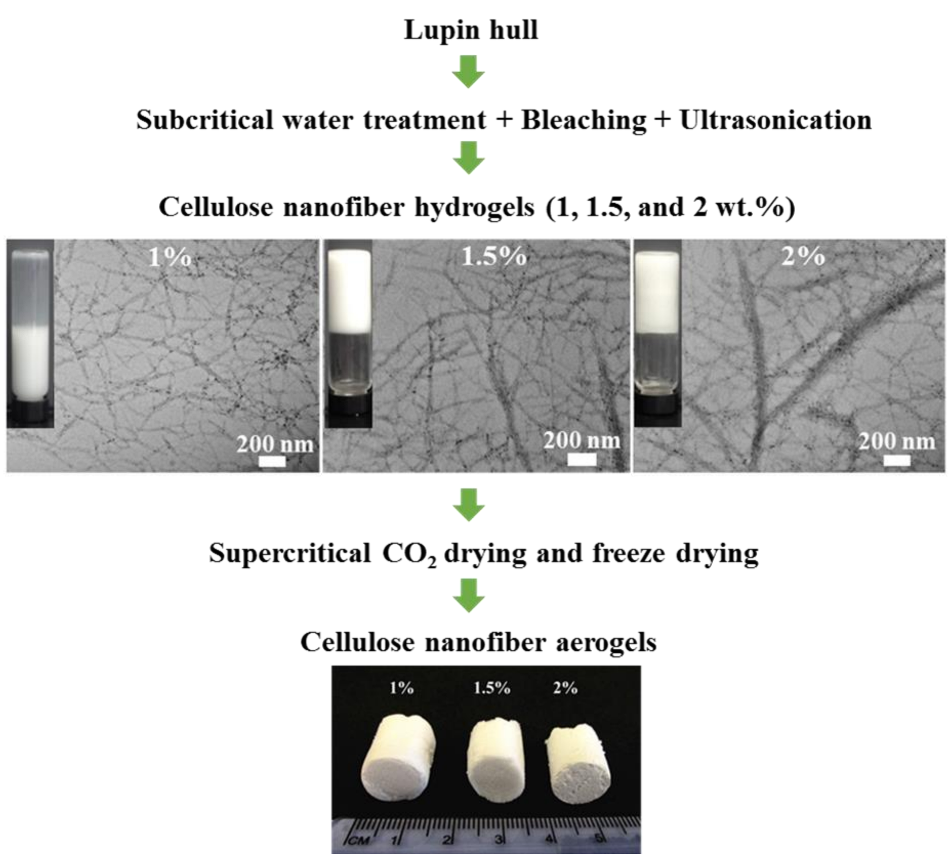
Graphical abstract



Highlights

- Lupin hull cellulose nanofiber aerogels were prepared by supercritical CO_2 drying.
- Highly porous (96.6-99.4%) and lightweight (0.009-0.05 g/cm^3) aerogels were formed.
- The resultant specific surface areas were in the range of 16-115 m^2/g.
- $SCCO_2$ drying was more effective in aerogel formation compared to freeze drying.
- All resultants aerogels had high crystallinity and thermal stability.

Abstract

In this study, aerogels were prepared from cellulose nanofiber hydrogels obtained via ultrasonication of subcritical water-assisted treated lupin hull. The $SCCO_2$ drying and freeze drying were evaluated for aerogel formation with initial hydrogel concentrations in the range of 1-2 wt%. The effects of concentration and drying method on resultant aerogel properties (density, porosity, specific surface area, pore size and pore volume), crystallinity, thermal behavior and morphology were investigated. The $SCCO_2$ drying was more advantageous in aerogel formation, allowing the production of aerogels with lower density (0.009-0.05 g/cm^3), and higher surface area (72-115 m^2/g) compared to freeze-dried aerogels at each concentration level investigated. The aerogel prepared from 1 wt% hydrogel concentration using $SCCO_2$ drying provided the lowest density of 0.009 g/cm^3, the highest porosity of 99% and the highest specific surface area of 115 m^2/g with high crystallinity index (72%) and thermal stability with degradation temperature of 310 °C.

Keywords: Aerogel; cellulose nanofiber; freeze drying; lupin hull; supercritical CO_2 drying.

1. Introduction

Aerogels are ultra lightweight and highly porous solid materials that have stimulated interest in a variety of applications, such as thermal/acoustic insulation, filtration, catalysis, and cushioning [1]. They are known for their extremely low densities

(0.0011 to ~0.5 g/cm^3), high porosity (> 80%), high specific surface area (up to 1000 m^2/g), low thermal conductivity (~15 mW/m K), low dielectric permittivity, and excellent shock absorption [2-5]. Aerogels obtained from cellulose nanofibers have received great attention due to renewability, biodegradability, and biocompatibility of cellulose, which is the most abundant biopolymer on Earth [6, 7]. Such materials offer new applications in the medical and pharmaceutical fields, where biocompatibility and biodegradability are needed [8], as well as in environmentally friendly packaging, and high performance and biodegradable nanocomposites [9]. Moreover, chemical functionality of resultant cellulose nanofiber aerogels creates more application areas, such as development of super hydrophobic materials via post-treatment with titanium dioxide nanoparticles [10] and silanes [11] to be used as oil adsorbents or separation medium for mixtures of oil/water liquids.

Aerogels exhibit increased ductility and flexibility arising from high aspect ratio and crystalline structure of nanofibers compared to traditional both organic and inorganic aerogels from derivatives of cellulose microfibers and metal oxides, which are usually brittle [9,12,13]. Moreover, chemical crosslinkers are not required in the gelation process. The aqueous gel is formed owing to the intramolecular hydrogen bonding and the long and entangled nanofiber network. ‘‘Sponge-like’’ aerogels are produced by replacing the liquid in the cellulose nanofiber gel by air via a suitable drying method that minimizes the collapsing of the network structure.

The first study on aerogel cellulose nanofiber was reported by Pääkkö et al. [9] who used two different freeze drying methods, cryogenic and vacuum, with 2 wt% aqueous cellulose nanofiber suspension, which was obtained from enzymatic treatment and fibrillation of softwood pulp. Using both methods, aerogels with low density around 0.02

g/cm^3 and high porosity up to 98% were obtained. Regarding the surface area, the cryogenic freeze drying yielded a value of 66 m^2/g, while the vacuum freeze drying yielded a lower value of 20 m^2/g. Similar freeze drying methods have been employed to obtain cellulose nanofiber aerogels from biomass such as wood pulp and poplar wood [8,14,15]. To reduce the extent of nanofiber aggregation during freeze drying due to ice sublimation, Sehaqui et al. [16] performed a solvent exchange of water to *tert*-butanol prior to freeze drying to obtain aerogels of wood pulp 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO)-oxidized cellulose nanofibers (1 wt% solid content in aqueous suspension). The resultant specific surface areas were as high as 153-284 m^2/g due to the exchange of water with *tert*-butanol. Therefore, *tert*-butanol was successfully employed in other studies to obtain cellulose nanofiber aerogels [17-19].

Supercritical drying has also been used to prepare aerogels as this method avoids the formation of surface tension and liquid-vapor interface in the aerogel wall [20]. Supercritical carbon dioxide ($SCCO_2$) has been the most appropriate fluid due to its mild critical pressure and temperature conditions (74 bar and 31 °C, respectively). However, there are few studies that reported production of cellulose nanofiber aerogels using $SCCO_2$ drying [10,16]. Korhonen et al. [10] obtained cellulose nanofiber aerogels from undried hardwood kraft pulp with 1.7 wt% aqueous suspension either by freeze drying in liquid nitrogen or liquid propane or by supercritical drying to prepare aerogel templates with the purpose of producing inorganic hollow nanotube aerogels by atomic layer deposition onto such templates. They reported that, in contrast to freeze drying, supercritical drying resulted in aerogels without major interfibrillar aggregation.

Drying the gel/hydrogel influences final aerogel properties, including density, porosity, specific surface area and morphology, in addition to the influence of raw material and concentration of the starting gel. In the case of cellulose nanofiber aerogel preparation, pretreatment/fibrillation methods and surface charge of cellulose nanofibers might be critical for most of the resultant aerogel properties. Thus, the aim of the present study was to investigate the effects of two different variables involved in the formation of aerogels, such as the drying method and concentration of the starting hydrogel. Lupin hull cellulose fibers obtained by combined subcritical water and bleaching treatment were used as the starting material. To the best of our knowledge, there is no research available on aerogel formation from cellulose nanofibers isolated from lupin hull or using the combined subcritical water-assisted treatment and high intensity ultrasonication, which is an environmentally friendly approach to reduce the use of hazardous chemicals and minimize waste generation. In this study, hydrogels with varying concentrations of cellulose fibers in the range of 1-2 wt% were prepared via ultrasonication. Two different drying processes, freeze drying and $SCCO_2$ drying, were examined for aerogel formation from cellulose nanofiber hydrogels. Aerogels properties, such as density, porosity, specific surface area, pore size, and morphology, were evaluated. Crystalline structure and thermal stability of the aerogels were also investigated.

2. Materials and methods

2.1. Materials

Lupin hulls were kindly provided by Ceapro Inc. (Edmonton, AB, Canada). Lupin hulls were ground to particle size < 1mm with a centrifugal mill (Retsch, Haan, Germany). The CO_2 (99.99% purity) was purchased from Matheson Tri-Gas Inc. (Montgomeryville, PA, USA). All chemicals used, such as sodium chlorite and ethanol, were of laboratory grade and obtained from Fisher Scientific (Pittsburgh, PA, USA).

2.2. Preparation of cellulose nanofiber hydrogels

Lupin hull cellulose fibers were used as the starting material to prepare cellulose nanofiber hydrogels. First, cellulose-rich residue was obtained using subcritical water technology at process conditions of 180 °C, 50 bar and 5 mL/min, as described in our previous study [21]. Then, the samples were bleached using 1.7% acidified sodium chlorite (50:1 v/v, suspension to acetic acid) with 10:1 v/w, liquid to solid ratio at 75 °C for 4 h according to a modified method [22] to obtain purified cellulose. Formation of hydrogels at various concentrations (1, 1.5 and 2 wt%) was induced by ultrasonication of lupin hull cellulose fibers in water for 40 min at 80% amplitude using an ultrasonicator (Model 705, 700 W, 50/60 Hz-Fischer Scientific, Pittsburgh, PA, USA). Ultrasonic treatments were carried out in an ice/water bath. Preliminary studies (data not shown) demonstrated that porous aerogels with high surface area were only obtained between 1 and 2 wt% lupin hull cellulose concentration. Hydrogels with concentrations below 1% did not exhibit a gel structure, also needed very long solvent exchange time (> 12 days) and collapsed during SC-CO_2 drying. Concentrations above 2 wt% resulted in the formation of less porous aerogels as observed from the SEM images due to aggregation of the cellulose nanofibers.

2.3. Preparation of cellulose nanofiber aerogels by freeze drying

Cellulose nanofiber hydrogels (7 mL) were placed in cylindrical tubes (6.0 cm height and 1.5 cm diameter) and frozen either by fast freezing with liquid nitrogen at -196 °C for 10 seconds or by slow freezing within a regular freezer at -18 °C for 24 h before the freeze-drying process. Then, the ice in the frozen hydrogel was sublimated at -45 °C and 15 Pa for 2 days using a freeze dryer (FreeZone, Labconco Corp., Kansas, MO, USA).

2.4. Preparation of cellulose nanofiber aerogels by $SCCO_2$ drying

Cellulose nanofiber aerogels were prepared from cellulose nanofiber hydrogels by first replacing the water in the hydrogels with ethanol via a multistage solvent exchange process, and then removing the ethanol from the alcogels using $SCCO_2$ drying. During the solvent exchange process, the hydrogels (7 mL) were soaked in 30, 50, 70, and 100% v/v excess ethanol (200 mL) for 1 h residence time, and 100% ethanol for 4 days, where ethanol was decanted and replaced with fresh ethanol every day to obtain the alcogel. The alcogels were then placed in cylindrical polypropylene molds to prepare alcogel monoliths. Monolith shape was preferred because hydrogels were placed in test tubes of cylindrical shape. Moreover, monolith shape allowed to make measurements to determine the density of the aerogels. Finally, the ethanol in the alcogel monoliths were removed by $SCCO_2$ drying to obtain cellulose nanofiber aerogels.

$SCCO_2$ drying was carried out in a laboratory scale $SCCO_2$ extraction system (SFT-110, Supercritical Fluids, Inc., DE, USA), similarly to the method used by Comin et al. [23] and Ubeyitogullari and Ciftci [24]. Briefly, first the alcogels were placed into a perforated (0.002 mm hole diameter) polypropylene basket (6.0 cm height and 1.5 cm diameter) that had a stainless steel frit on top of the perforated bottom. The basket was then placed into the high pressure vessel, containing the alcogels that were placed on the frit. Then, an excess amount of ethanol (20 mL) was poured into the basket to account for the loss of ethanol from the alcogel through evaporation before the set pressure and temperature were reached. Temperature of the vessel was set to 40 °C using the temperature controller. After the set temperature was reached, the system was pressurized to 100 bar and kept at the set pressure for 10 min using the double head syringe pump (model 260D, Teledyne ISCO, Lincoln, NE, USA). Then, the shut off valve was opened and the CO_2 flow rate was set to 0.5 L/min (measured at ambient conditions) and maintained constant using the micrometering valve. The CO_2 flow rate was measured by a gas flow meter. $SCCO_2$ drying was performed at the set pressure, temperature, and CO_2 flow rate for 4 h. After that time, the system was depressurized at the same CO_2 flow rate and temperature, and the samples were collected from the vessel and stored at room temperature (21 °C) for further characterization.

2.5. Characterization of the aerogels

2.5.1. Bulk density and porosity

Volume of the aerogel monolith was determined from its final dried dimensions that were measured using a caliper with a precision of 0.05 mm, and the weight of aerogel monolith was determined using a sensitive electronic balance with a precision of 0.1 mg. Then, the bulk density (d_a) values were obtained by calculating the ratio of mass to volume.

The porosity (P) of the aerogels was calculated using the d_a values in the Eq. (1), where the density of the crystalline cellulose nanofibers (d_n) is equal to 1.6 g/cm^3 [14].

$$P\ (\%) = (1 - \frac{d_a}{d_n}) * 100 \qquad \text{(Eq. 1)}$$

2.5.2. Specific surface area and pore size

Brunauer–Emmett–Teller (BET) surface area and Barrett-Joyner-Halenda (BJH) pore size of the cellulose nanofiber aerogels were determined by low-temperature nitrogen adsorption-desorption method (ASAP 2460, Micromeritics Instrument Corporation, Norcross, GA, USA). Small pieces of aerogel samples (0.1-0.3 g) were placed in the sample tube, and then degassed under vacuum at 115 °C for 10 h prior to analysis. Nitrogen sorption experiments were performed at -196 °C. Specific surface area was determined at a relative pressure (p/p_0, equilibrium pressure of nitrogen at the sample surface/saturation pressure of nitrogen) between 0.05 and 0.3 by multipoint BET adsorption characteristics. Pore size distribution was at $p/p_0 > 0.35$. The BJH average pore size and pore volume were calculated considering pores between 1.7 nm and 300 nm width based on the Kelvin equation, which assumes that the pores are rigid and all of the same simple shape (cylinders or parallel-sided slits). However, porosity was calculated considering all pores, especially pores larger than 300 nm [25].

2.5.3. Microscopic analysis

The morphology of the aerogel samples was analyzed by a field emission scanning electron microscope (FE-SEM) (S4700 FE-SEM, Hitachi, Tokyo, Japan) at 2 kV and 15 mA under low vacuum mode. Cross-sections (1 mm thickness) were cut from the aerogel monoliths and placed on aluminum SEM specimen stubs with double-side conductive carbon tape. Prior to analysis, the samples were sputter-coated with chromium (Desk V HP TSC, Denton Vacuum LLC, NJ, USA).

The morphology of the cellulose nanofiber hydrogels were analyzed by a transmission electron microscope (TEM) (H7500 TEM, Hitachi, Tokyo, Japan) operated at 80 kV. One drop of aqueous nanofiber hydrogel (~0.1 g cellulose nanofiber/mL) was placed onto carbon-coated TEM grids and the sample was then negatively stained with 1% uranyl acetate and allowed to dry under ambient conditions. For a typical TEM analysis, the solvent is first evaporated. Diameters of cellulose nanofibers were calculated using ImageJ processing software (IJ1.46) by loading the TEM images into the software and measuring the fiber diameters. Scale bars on each TEM image were used for calibration of the software. Approximately 100 measurements were done using 10 TEM images. The average diameters and size distributions were determined by drawing straight lines from the selected fibers in the corresponding TEM images.

2.5.4. X-ray diffraction analysis

Crystallinity of the aerogel samples was determined using a PANalytical Empyrean Diffractometer (Empyrean, PANalytical B.V., Almelo, Netherlands) operated at 45 kV, 40 mA with Cu Kα beam monochromator, and equipped with PIXcel^{3D} detector that was operated with 1D detection. The ground aerogel samples were spread on the sample holder and spun continuously at the rate of 3.75 rpm throughout the analysis. The samples were scanned at a scanning speed of 1.267 °/min with a step size of 0.05° within the range of 2°-40° (2θ).

The crystallinity index (CI) was calculated from the heights of the intensity of the crystalline region (I_{002} at 2θ = 22.5º) and the intensity of the amorphous region (I_{am} at 2θ = 18.5º) using the Segal method [26]:

$$CI\% = \frac{I_{002}+I_{am}}{I_{002}} * 100 \qquad \text{(Eq. 2)}$$

2.5.6. Thermo-gravimetric analysis

Thermo-gravimetric (TG) analyses were done using a TG 209 F1 Libra TG analyzer (TG 209 F1 Libra, NETZSCH, Selb, Germany). An aerogel sample (5-10 mg) was placed in a sealed aluminum pan and heated from room temperature to 600 °C at a heating rate of 10 °C/min under nitrogen flow at 20 mL/min.

2.6. Statistical analysis

Data were presented as mean ± standard deviation based on at least duplicates or triplicates. Statistical analyses were performed using the SPSS (version 17.0) software package at 95% confidence interval.

3. Results and discussion

3.1. Aerogel properties

Lightweight and white sponge-like aerogels were obtained without significant collapse upon complete removal of water through freeze drying and ethanol removal through $SCCO_2$ drying. Table 1 shows the properties of the cellulose nanofiber aerogels prepared by those drying techniques from cellulose nanofiber hydrogels at different concentrations ranging from 1 to 2 wt% (initial solid cellulose nanofiber content in the hydrogel). The resultant aerogel properties were greatly affected by the initial hydrogel concentration and the drying method employed as shown in Table 1. The $SCCO_2$ drying resulted in 'better' aerogel properties, such as lower density, and higher specific surface area compared to those of freeze-dried aerogels at each concentration level investigated. Decreasing hydrogel concentration regardless of the drying method employed resulted in decreasing of density, and increasing of porosity and specific surface area during aerogel formation.

The lowest density of 0.009 g/cm^3 was achieved at 1 wt% hydrogel concentration with $SCCO_2$ drying, whereas freeze-dried aerogel yielded a density of 0.023 g/cm^3 at the same concentration due to pore collapsing during ice sublimation. The density of the aerogel is negatively correlated with the porosity as shown in Eq. (1). As a result of such relationship, the highest porosity was calculated as 99.4% for the $SCCO_2$-dried aerogel, having the lowest density of 0.009 g/cm^3.

The lowest porosity was calculated as 96.6% for the freeze-dried aerogel which had the highest density of 0.054 g/cm^3 at 2 wt% concentration. On the other hand, concentration and drying method had no significant effect on average pore size and pore volume of the aerogels, with values in the range of 7.1-11.7 nm, and 0.08-0.36 cm^3/g, respectively. Similar pore size is expected as pores are mainly formed during gelation. However, shrinkage or collapse of small pores during freeze drying results in formation of larger pores, which are above the BJH's method working limit. The BJH average pore size was calculated for the pores in the range of 1.7-300 nm, which is better than using the BET method for pores up to 200 nm (Heath & Thielemans, 2010).

The BET surface area (specific surface area determined using the BET method) of freeze-dried aerogels slightly increased from 16 to 20 m^2/g with decreasing concentration from 2 to 1 wt%, while the concentration increase had more pronounced effect on the BET surface area of $SCCO_2$-dried aerogels with values increasing from 72 to 115 m^2/g as the concentration decreased from 2 to 1 wt%.

A similar phenomenon where increasing hydrogel concentration and density led to a lower BET surface area of the resultant aerogel was also observed for aerogel formation based on cellulose nanofibers [8,9], or cellulose [27]. Increasing hydrogel concentration per unit volume results in an increased density with less porous structure and therefore resulted in less surface area. Aulin et al. [8] prepared cellulose nanofiber aerogels at 0.0031-3.13 wt% of carboxymethylated nanofibers of dissolved pulp. They reported that aerogels obtained from 2 wt% hydrogel resulted in a density of 0.02 g/cm^3 and BET area of 15 m^2/g, whereas increasing the concentration to 3.13 wt% yielded a higher density of 0.030 g cm^{-3} with a lower BET area of 11 m^2/g due to the presence of more nanofiber per unit volume, which results in a denser structure. Comparison of the present aerogel BET surface area values with the previous aerogel studies based on cellulose nanofibers in the similar concentration range tested in this study showed that the BET surface area was affected by many parameters in addition to concentration, such as freezing type/speed and additional solvent exchange time and solvent. The values for BET areas of cellulose nanofiber freeze-dried aerogels (2 wt%) obtained in this study were similar to those of vacuum freeze-dried aerogels reported by Pääkkö et al. [9], who obtained the area of 20 m^2/g using 2 wt% cellulose nanofiber gel of softwood pulp. The reason for the small surface area of freeze-dried aerogels (16-20 m^2/g) in this study can be explained by the use of the BJH pore volume (0.08-0.09 cm^3/g) calculated in the range of 1.7-300 nm, and also collapse of the pores during freeze drying. On the other hand, BET surface area values of this study were lower than those obtained by Sehaqui et al. [16], who conducted additional solvent exchange step with *tert*-butanol prior to freeze drying, achieving 153-284 m^2/g with 1 wt% concentration of wood pulp cellulose nanofiber hydrogel. However, the use of *tert*-butanol in the solvent exchange step for the preparation of aerogels from coconut shell cellulose nanofiber gel (0.5 wt%) resulted in much lower BET surface area (9.1 m^2/g) [19], which could be related to the influence of the other variables involved in the formation of aerogels, such as source, treatment/mechanical fibrillation method used or surface charge of the starting cellulose nanofibers. Surface modification pretreatments of cellulose fibers to add ionic groups on the surface by oxidation and carboxymethylation before mechanical treatments can affect the final aerogel properties, resulting in high specific surface areas.

3.2. Morphology

3.2.1. Cellulose nanofiber hydrogels

Aerogel properties might vary depending on the initial hydrogel characteristics of the cellulose nanofibers. In this study, ultrasonication of cellulose fibers in water form hydrogel structures via formation of nanofiber network. As native cellulose has highly hydrophilic nature, cellulose nanofibers are able to retain a large amount of water [28]. Thus, aqueous suspensions of cellulose nanofibers show gel-like properties because of the formation of an entangled network structure attributed to the high aspect ratio and the high specific surface area of the nanofibers [29,30]. Besides, a chemical cross-linker addition was not required to induce gelation of the present hydrogels, which were formed by the physical cross-linking of the cellulose nanofibers.

The TEM images of hydrogels displayed a classical web-like network structure, revealing long entangled nanofiber filaments with diameters of 5-100 nm, and the lengths in several microns which is challenging to measure (Fig. 1). As a result of the physical cross-linking due to more inter- and intra-molecular hydrogen bonds and entanglements, formation of white, sponge-like aerogels with well-defined shapes without significant shrinkages were observed (Fig. 1).

Flow ability of the hydrogels varied with changing concentration from 1 to 2 wt%. A partial gelatinous structure was observed in the 1 wt% hydrogel; however, increasing the concentration above 1 wt% resulted in viscous hydrogels with increased aggregation of nanofibers and no flow ability (inset pictures, Fig. 1). It was reported by Chen et al. [14] that increasing cellulose nanofiber content from 0.1 to 1.5 wt% resulted in increased strength of the formed aerogels due to the increased degree of physical cross-linking caused by more inter- and intra-molecular hydrogen bonds and entanglements in the viscous hydrogels. They evaluated the effect of hydrogel concentration obtained from wood cellulose nanofibers on water uptake capability of the resultant aerogels and reported that water uptake capability (ratio of water to aerogel) of aerogels decreased from 155 to 54 with increasing concentration from 0.1 to 1.5 wt% due to increase of such physical cross-linking.

3.2.2. $SCCO_2$-dried aerogels

The morphologies of $SCCO_2$-dried aerogels as a function of the initial hydrogel concentration are presented in Fig. 2. Using $SCCO_2$ drying (100 bar, 40 °C, 0.5 L/min CO_2 flow rate, measured at ambient conditions) to prepare aerogels was effective to form highly porous structures. These $SCCO_2$ drying conditions were based on previous conditions used for formation of aerogels [23, 24], and preliminary studies for the lowest pressure and temperature for efficient ethanol removal from the alcogel.

The morphologies with three-dimensional open nanoporous network structure of entangled nanofibers are shown in the SEM images in Fig. 2. At 1 wt% concentration, more homogenous and porous aerogel network structure was seen, which was built up with individual cellulose nanofibers without any visible or occasional aggregation of fibers (Fig. 2a-c). However, random formation of bigger fiber bundles and aggregates were observed (Fig. 2d-i) as the concentration was increased from 1 to 2 wt%, which was the result of increased attraction among nanofibers due to increased hydrogen bondings.

3.2.3. Freeze-dried aerogels

Morphologies of freeze-dried aerogels, initially frozen with liquid nitrogen, were different than those of $SCCO_2$-dried aerogels as shown in Fig. 3. As expected, the $SCCO_2$

drying method was more effective in forming aerogels, exhibiting more homogenous porous structures that were composed of thinner and not aggregated individualized fibers, which is attributed to the lack of any intermediate liquid-vapor interface and no surface tension in the gel pores [20].

Fig. 3a-b showed cellulose nanofibers in the 1 wt% hydrogel self-assembled into long fibers in the longitudinal direction during the freeze-drying process. When the hydrogel concentration exceeded 1 wt%, the number of pores decreased accompanied with the formation of more sheet-like structures. Even though the average pore size was found to be in the range of 9.4-10.9 nm as reported in Table 1, formation of micrometer-sized pores were identified on the sheet like surfaces (Fig. 3d and 3f).

Effect of initial freezing stage

Freeze drying of cellulose nanofiber hydrogels involves two major stages. The initial stage is freezing of the hydrogels into ice crystals with nanofibers trapped among them, leading to the formation of an ordered structure, and the final stage is the sublimation of ice, resulting in the formation of a porous network structure [31]. Freezing speed at the initial stage of freeze drying plays an important role, affecting the microstructure properties of the formed aerogels, which can be related with the size and distribution of the ice crystals formed during freezing [32]. Fig. 4 compares the morphologies of the aerogels obtained with the two different freezing methods at the initial stage of freeze drying. Hydrogels of 1 wt% were fast frozen with liquid nitrogen at a temperature of -196 °C, and slow frozen within a regular freezer at a temperature of -18 °C. The use of fast and slow freezing methods prior to sublimation resulted in aerogels with different structures. Using fast freezing with liquid nitrogen was more effective in preserving the porous network structure (Fig. 4a-b). More compact two-dimensional sheet-like structures were formed, displaying wave-like roughness during the relatively slow freezing process at -18 °C (Fig. 4c-d).

3.3. Crystallinity

Fig. 5 shows the comparison of the X-ray diffraction patterns of the purified cellulose and $SCCO_2$-dried aerogels as a function of the initial hydrogel concentration. Crystalline structure arrangement of the native cellulose was maintained in all aerogels after applied treatments (subcritical water treatment, ultrasonication and drying) as all the patterns displayed a typical cellulose I structure with three well-defined crystalline peaks around 2θ = 16.5°, 22.5° and 34.5° [33]. However, the CI value of the 1 wt% (72.4%) aerogel was found to be slightly lower than that of the 1.5 wt% aerogel (73.7%) and the 2 wt% aerogel (75.1%), but it was not statistically significantly different ($p>0.05$). Such a decrease in crystallinity might be attributed to the damaging effect of the ultrasonication process at increased intensity (increased time and/or amplitude).

Li et al. [34] reported a significant decrease in crystallinity of microcrystalline cellulose from 82% to 73% after ultrasonication at 1500 W for 15 min to obtain cellulose nanocrystals. Even though all the aqueous cellulose fibers were processed for 40 min, the hydrogels with concentrations of 1.5 and 2 wt% might not be adequately agitated as the one with 1 wt% due to increased viscosity and, thus decreased flow ability during ultrasonication. Furthermore, the resultant cellulose nanofiber aerogels still had high crystallinity index with a value above 70%, which is crucial in terms of industrial applications like insulation as high crystallinity is closely related with high thermal stability and good mechanical properties.

3.4. Thermal stability

Investigation of thermal stability is important to evaluate the capability of cellulose nanofiber aerogels for high-temperature applications like thermal insulation. Fig. 6 presents the TG curves of subcritical water treated and bleached lupin hull cellulose and the $SCCO_2$-dried cellulose nanofiber aerogels at different concentrations of 1-2 wt%. All $SCCO_2$-dried aerogels displayed a decomposition behavior that is highly similar to that of the purified cellulose fibers with the onset of degradation temperature occurring at approximately 310 °C. A decrease of the onset degradation temperature would be expected when the crystalline region is damaged, which could lead to a decrease in thermal stability [35]. There was not such a decrease in the thermal stability, indicating that the above-mentioned crystallinity decrease of 1 wt% aerogel was not significant enough to affect the thermal stability.

4. Conclusions

Highly porous and lightweight cellulose nanofiber aerogels were successfully prepared using $SCCO_2$ drying and freeze drying. The resulting aerogels had the highest specific surface area of 115 m^2/g, the highest porosity of 99% and the lowest density of 0.009 g/cm^3 via $SCCO_2$ drying of 1 wt% hydrogel with three-dimensional open nanoporous network structures. Increasing concentrations from 1 to 2% led to more aggregated structures and large fiber bundles after $SCCO_2$ drying and liquid nitrogen freeze drying, while formation of two-dimensional sheet-like structures were observed during regular freeze drying. All resultants aerogels had high crystallinity (>72%) and thermal stability (thermal degradation temperature of 310 °C). Aerogels obtained from renewable and biodegradable cellulose nanofibers can be used in various applications, including environmentally friendly food packaging and tissue engineering scaffolds.

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Tables and Figures

Property	**$SCCO_2$ drying**			**Freeze drying**		
	1 wt%	**1.5 wt%**	**2 wt%**	**1 wt%**	**1.5 wt%**	**2 wt%**
Density (g/cm^3)	0.009[a]* (0.001)**	0.019[b] (0.001)	0.050[c] (0.001)	0.023[a] (0.001)	0.030[b] (0.001)	0.054[c] (0.001)
Porosity (%)	99.4[a] (0.3)	98.8[b] (0.5)	96.9[c] (0.5)	98.6[a] (0.3)	98.1[a] (0.5)	96.6[b] (0.3)
BET surface area (m^2/g)	115[a] (4)	93[b] (7)	72[c] (5)	20[a] (1)	18[b] (1)	16[c] (2)
BJH pore size (nm)	8[a] (2.2)	11.7[b] (1.7)	7.1[c] (2.1)	10.8[a] (1.8)	10.9[a] (1.4)	9.4[b] (1.9)
BJH pore volume (cm^3/g)	0.32[a] (0.03)	0.36[a] (0.02)	0.17[b] (0.05)	0.08[a] (0.01)	0.09[a] (0.02)	0.09[a] (0.02)

Table 1
Comparison of cellulose nanofiber aerogel properties obtained via different drying methods.

*Different letters in the same row within each drying method are significantly different at $p<0.05$.
**Data values in parenthesis are standard errors.
BET: Brunauer–Emmett–Teller; BJH: Barrett-Joyner-Halenda.

Figure Captions

Fig. 1. TEM images of cellulose nanofiber hydrogels (1, 1.5, and 2 wt%) and their aerogel appearance at macroscopic level.

Fig. 2. Morphologies of $SCCO_2$-dried aerogels as a function of the initial hydrogel concentration ((a-c) 1 wt%, (d-f) 1.5 wt%, and (g-i) 2 wt%).

Fig. 3. Morphologies of freeze-dried aerogels as a function of the initial hydrogel concentration ((a and b) 1 wt%, (c and d) 1.5 wt%, and (e and f) 2 wt%).

Fig. 4. Morphologies of freeze-dried aerogels using fast freezing with liquid nitrogen (a and b) and slow freezing within a regular freezer (c and d) at a hydrogel concentration of 1 wt%.

Fig. 5. XRD patterns of aerogels at different concentrations.

Fig. 6. TG curves of aerogels at different concentrations.

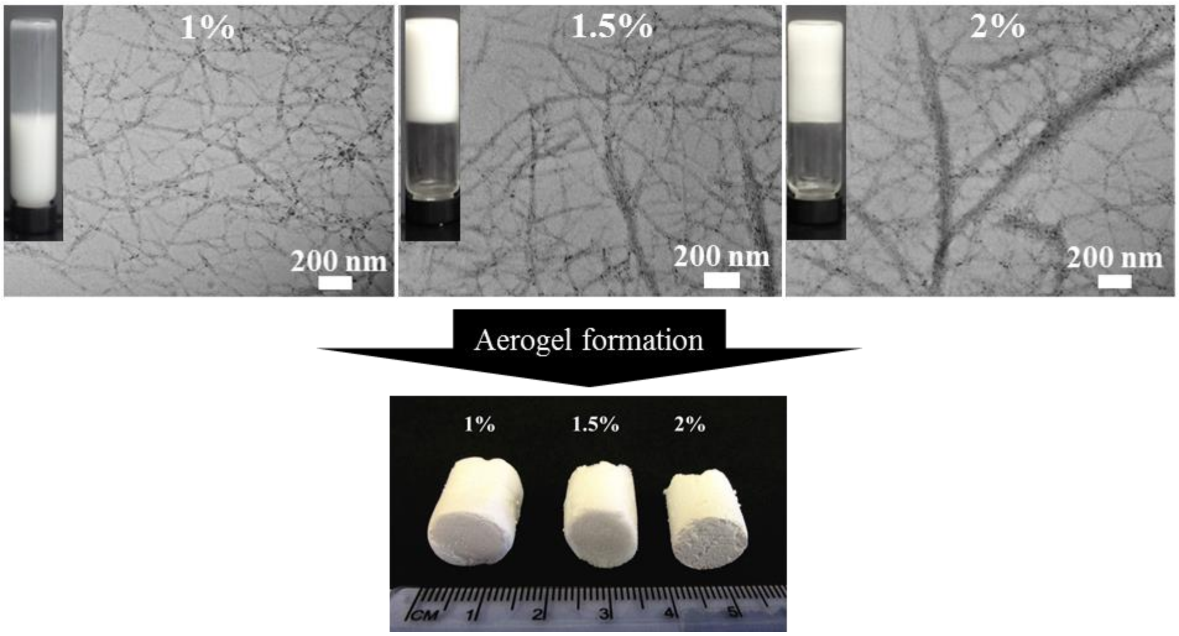

Fig. 1

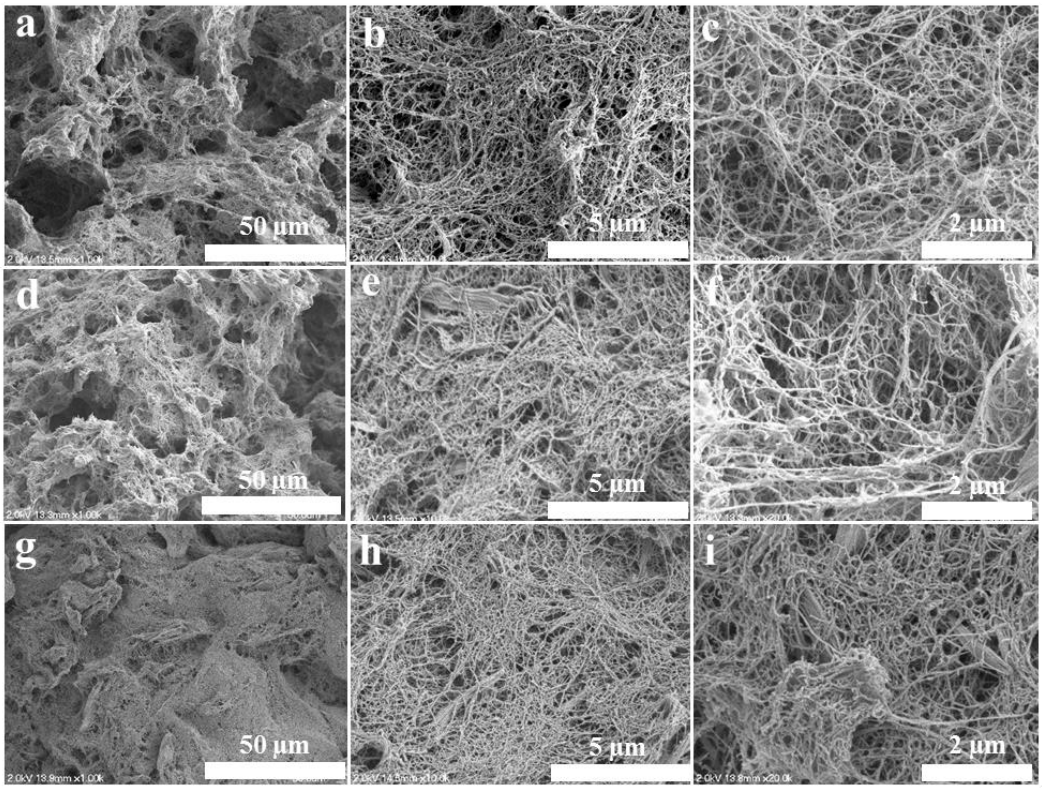

Fig. 2

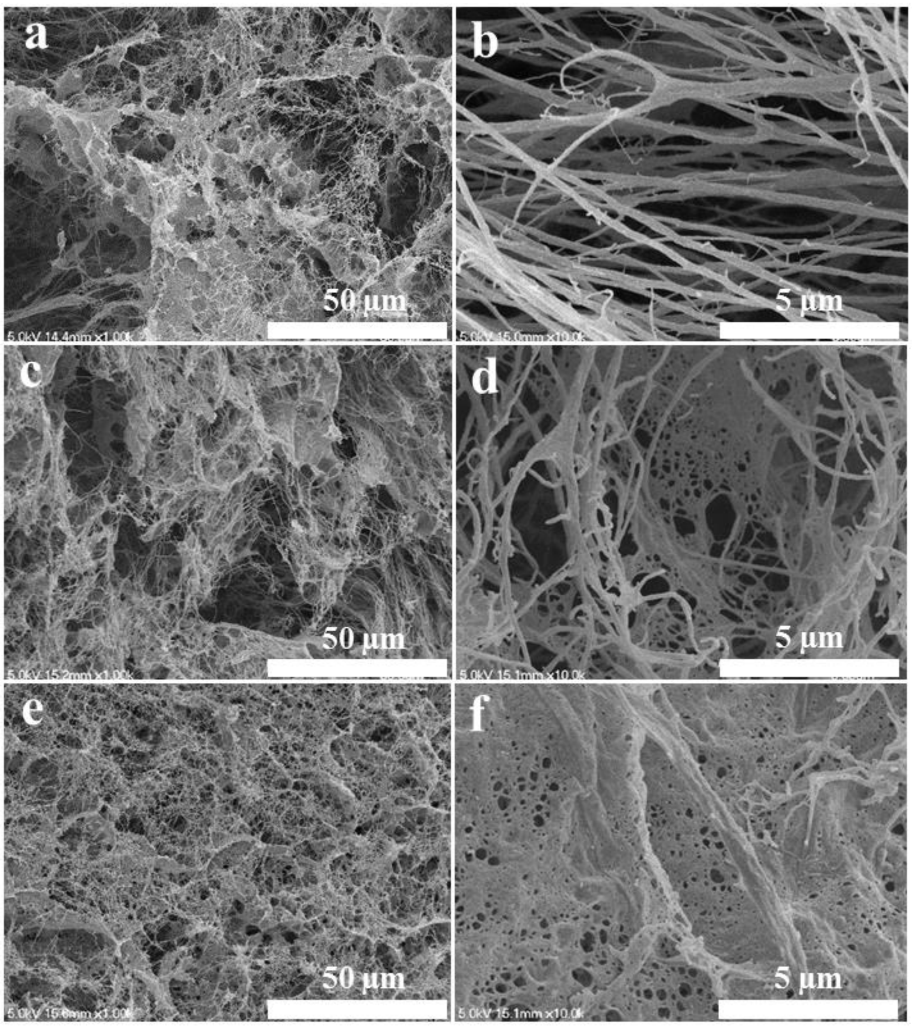

Fig. 3

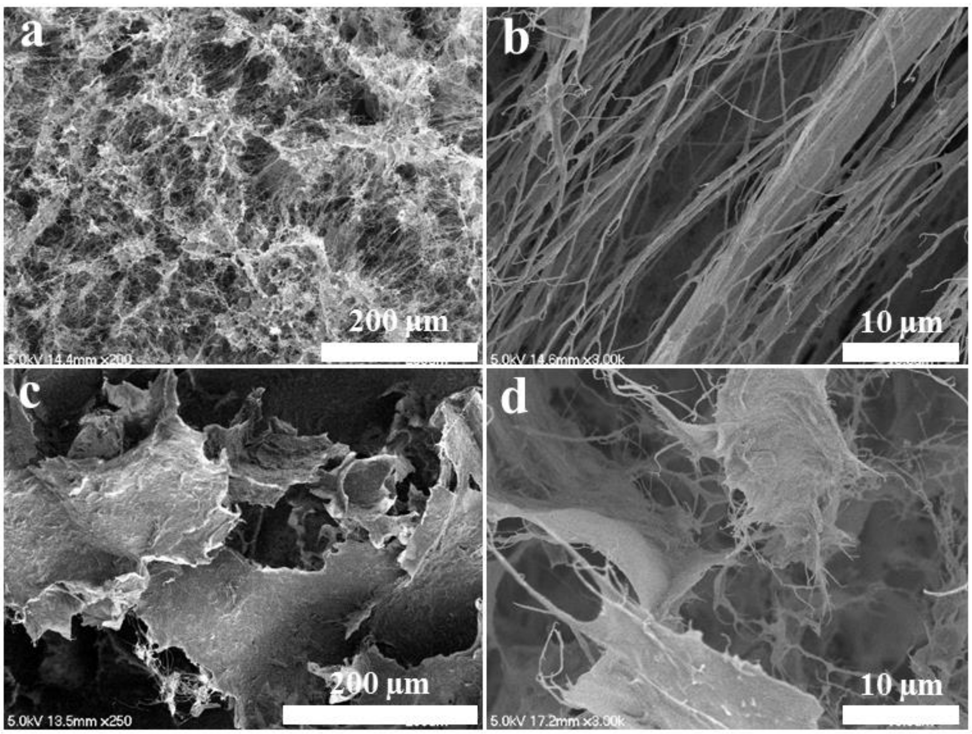
a
200 µm
5.0kV 14.4mm x200
b
10 µm
5.0kV 14.6mm x3.00k
c
200 µm
5.0kV 13.5mm x250
d
10 µm
5.0kV 17.2mm x3.00k

Fig. 4

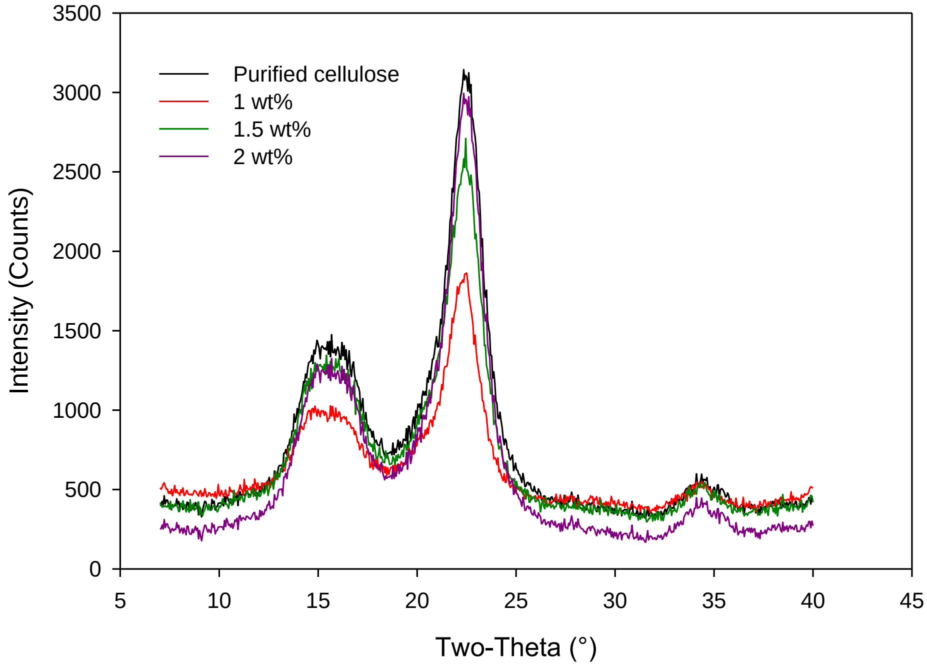
3500
3000
2500
2000
1500
1000
500
0
Intensity (Counts)
5
10
15
20
25
30
35
40
45
Two-Theta (°)
Purified cellulose
1 wt%
1.5 wt%
2 wt%

Fig. 5

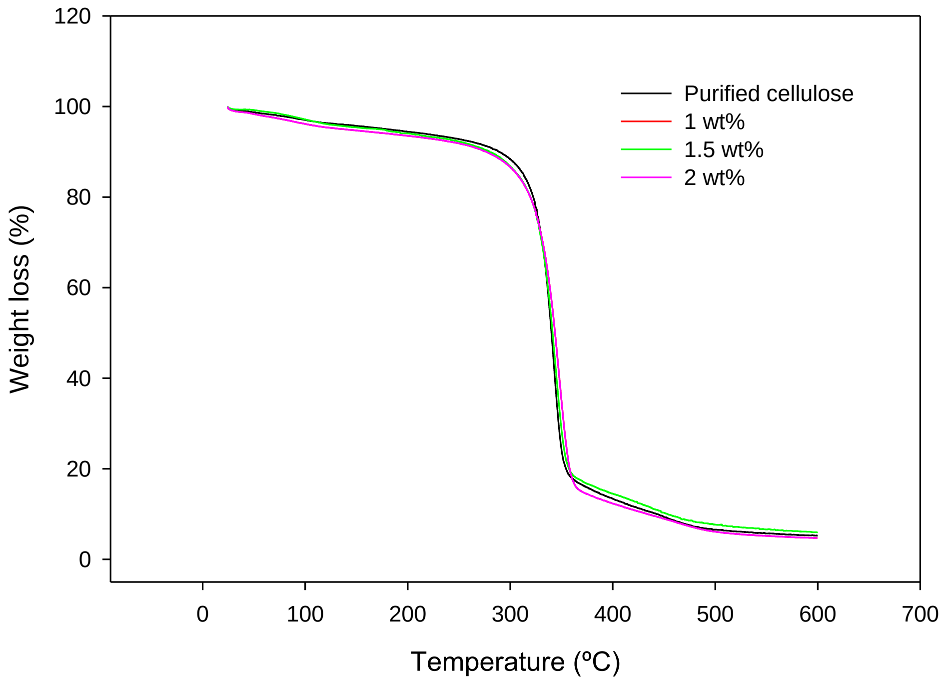
120
100
80
60
40
20
0
Weight loss (%)
0
100
200
300
400
500
600
700
Temperature (ºC)
Purified cellulose
1 wt%
1.5 wt%
2 wt%

Fig. 6